

Yuchen Wang

Agentic post-training | ML systems

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Education

University of Illinois Urbana-Champaign

Master of Computer Science

2025 – 2027 (expected)

Champaign, IL

Peking University

B.S. in Intelligent Science and Technology (AI); Zhi Class (Elite Program), Top 10%

2021 – 2025

Beijing, China

Internships

Alibaba U.S. – Accio Team

May 2026 – Dec 2026

Research Intern, LLM Training Algorithms & Infrastructure

Sunnyvale, CA

- Co-developed core RL post-training algorithms and engineered execution infrastructure for **Occamy-1.0 [arXiv]**, a 35B-A3B agent with **500K+ 🤖 downloads across Occamy variants**: captured exact model-boundary token IDs and verifier outcomes across OpenClaw, Hermes, and Accio Work.
- Recovered long-horizon SFT/RL supervision through workspace replay and context–action–reward realignment after tool failures and compaction, keeping trajectories usable for post-training.
- Shipped Occamy with Accio; versus its Qwen3.6 base, the released model raised AutomationBench strict pass **7.5% → 27.6%** and Claw-Eval average **69.5 → 82.2**, while reducing Claw-Eval T/C full-trial time **36.6%**.

Freedom Technology

May 2025 – Aug 2025

Research Intern, 3D Vision and Generative AI

Beijing, China

- Raised point-cloud geometric fidelity from **32.7% to 85.3%** by conditioning ControlNet on depth and surface normals while preserving edges and planar surfaces.

Open Source

- Halved 120B MoE TP2 BF16 conversion peak (**7.88 → 3.94 GiB**) in **vLLM**; fixed calibrated FP8 KV prefill in **FlashInfer**.
- Preserved source sharding in **HF Datasets** streaming shuffle, restoring four-worker use on 19 shards; made packed QSA selection document-local in **mcCore-bridge**, **2.70× faster** on an 8K synthetic selector case.
- Developing **Bridge-local Qwen4-Exp** support for QSA, per-layer embeddings, gated residuals, and HF conversion.
- Bypassed the Ray driver for **NeMo RL** top-*k* teacher transfers (**4.44–5.31×** on 8×B200); added **GDN kernel-fused Q/K normalization** and **selective Mamba recompute** in Megatron-LM.
- Fixed silent small-gradient failure in **Muon**; proposed **checkpoint-safe Adam counters** for Megatron-LM; co-authored a **NeMo Gym** fix preserving HTTP errors.

Research

RareDx: Graph-Grounded RL for Rare-Disease Diagnosis [arXiv]

Jun 2026 – Sep 2026

Advisor: Prof. Tianyu Liu, Tsinghua University

- Co-designed RL rewards for partial diagnostic matches using disease-ontology distance, biomedical similarity, and phenotype–disease links. With SFT and retrieval, 27B diagnosis Hit@10 increased from **28.53% to 40.76%**.

CineFlow: Dependency-Driven Parallel Video Generation [Paper]

Oct 2025 – May 2026

Advisor: Prof. Fan Lai, University of Illinois Urbana-Champaign

- Developed a training-free, critical-path scheduler for video diffusion using semantic DAGs, trajectory fusion, and online edge pruning; analyzed 1K+ MovieGenBench prompts.
- Delivered **1.7–5.5× end-to-end speedup**, 1.30–2.02× lower P90 latency, and **5.4–17.3% higher VBench overall** across Wan2.2-5B, CogVideoX-5B, and HunyuanVideo on 8×H100.

Dynamic Prefill Optimization via Adaptive Online Packing [Report]

Aug 2025 – Jan 2026

University of Illinois Urbana-Champaign

- Built an AIMD batch-size controller driven by live p95 TTFT and packed mixed-length prompts with greedy/DP scheduling; cut TTFT by **up to 20%** on production-style DynamoLLM traces.

FlashAttention-style CUDA Kernels for GPT-2 [Report]

Aug 2025 – Dec 2025

University of Illinois Urbana-Champaign

- Accelerated GPT-2 through tiled attention, online softmax, and kernel fusion, raising arithmetic intensity from **15 to 157 FLOP/byte**, cutting HBM traffic roughly **10×**, and improving end-to-end speed up to 9%.

Reinforcement Learning for Legal Reasoning [Thesis]

Jan 2025 – Mar 2025

Advisor: Prof. Yansong Feng, Peking University

- Post-trained a legal model with Zero-RL, chain-of-thought-distilled SFT, and GRPO; reached **57.6% accuracy**, beating SFT-only and RL-only baselines.

Service, Awards & Technical Skills

Reviewer, WSDM 2027

Awards: Merit Student; Zhi Class Scholarship; CMO (Provincial First Prize), 2020.

Languages: Python, C++, CUDA. **Systems:** PyTorch, Ray, FSDP, Megatron-LM, NeMo RL, vLLM, SGLang, Nsight.

Methods: RL, distributed training, diffusion transformers, ControlNet, 3D vision and point clouds.